

Problem

In the circuit of Fig. 7.93,

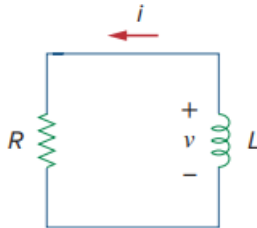
$$v(t) = 80e^{-10^3 t} \text{ V}, \quad t > 0$$

$$i(t) = 5e^{-10^3 t} \text{ mA}, \quad t > 0$$

(a) Find R , L , and τ .

(b) Calculate the energy dissipated in the resistance for $0 < t < 0.5 \text{ ms}$.

Fig. 7.93:



Step-by-step solution

Step 1 of 6

Consider the following circuit diagram:

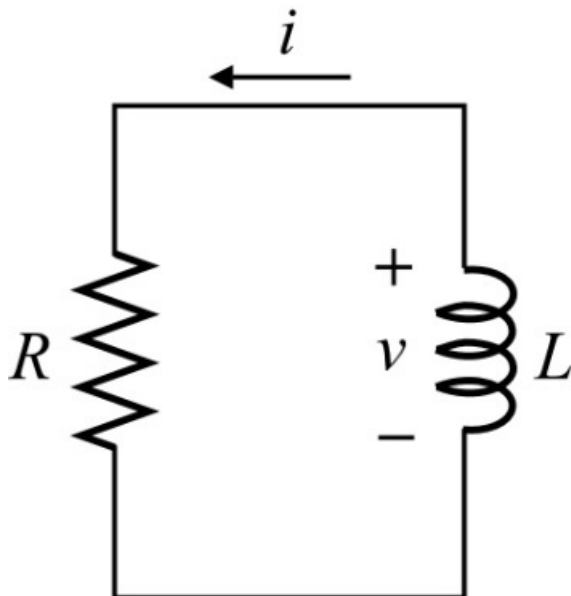


Figure 1

Step 2 of 6

(a)

The current through the inductor, $i(t) = 5e^{-10^3 t}$ mA (1)

The voltage across the inductor, $v(t) = 80e^{-10^3 t}$ V

Write the general expression for current through the inductor.

$$i(t) = i_0 e^{-\frac{t}{\tau}} \text{ A} \dots\dots (2)$$

Compare equation (1) and equation (2).

$$\tau = \frac{1}{10^3} \text{ s}$$

Therefore, the value of time constant, τ is $\boxed{\frac{1}{10^3} \text{ s}}$.

Step 3 of 6

Apply Kirchhoff's voltage law to circuit shown in Figure 1.

$$Ri - v = 0$$

$$v = Ri$$

Substitute $5e^{-10^3 t}$ mA for $i(t)$ and $80e^{-10^3 t}$ V for $v(t)$ in the above equation.

$$80e^{-10^3 t} = R(5 \times 10^{-3})e^{-10^3 t}$$

$$R = \frac{80e^{-10^3 t}}{(5 \times 10^{-3})e^{-10^3 t}}$$

$$= 16 \times 10^3$$

$$= 16 \text{ k}\Omega$$

Step 4 of 6

Therefore, the value of resistor, R is $\boxed{16 \text{ k}\Omega}$.

Step 5 of 6

Write an expression for time constant.

$$\tau = \frac{L}{R_{\text{eq}}}$$

$$L = \tau R_{\text{eq}}$$

Substitute $16 \text{ k}\Omega$ for R and $\frac{1}{10^3}$ for τ in the above equation.

$$L = \frac{16 \times 10^3}{10^3}$$

$$= 16 \text{ H}$$

Hence, the value of inductor, L is $\boxed{16 \text{ H}}$.

Step 6 of 6

(b)

Write an expression for energy dissipated in the resistance for $0 < t < 0.5 \text{ ms}$.

$$\begin{aligned}w_R(t) &= \int_0^t i^2 R dt \\&= \int_0^{0.5 \times 10^{-3}} \left(5 \times 10^{-3} e^{-10^3 t}\right)^2 (16 \times 10^3) dt \\&= \int_0^{0.5 \times 10^{-3}} 0.4 e^{-2 \times 10^3 t} dt \\&= \frac{0.4 e^{-2 \times 10^3 t}}{-2 \times 10^3} \Big|_0^{0.5 \times 10^{-3}} \\&= \frac{0.4 e^{-1} - 0.4}{-2 \times 10^3} \\&= \frac{(0.4)(0.368) - 0.4}{-2 \times 10^3} \\&= \frac{0.147 - 0.4}{-2 \times 10^3} \\&= \frac{-0.253}{-2 \times 10^3} \\&= 126.5 \mu\text{J}\end{aligned}$$

Therefore, the energy dissipated in the resistance, $w_R(t)$ for $0 \leq t \leq 0.5 \text{ ms}$ is $\boxed{126.5 \mu\text{J}}$.